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ACADS ACLY ACO2 ACOX1 ADCY6 ADIG ADIPOQ
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APOE ARAF ARL4A ATL2 ATP1B3 ATP5PO BAZ2A BCKDHA
BCL2L13 BCL6 C3 CAT CAVIN1 CAVIN2 CCNG2 CD151
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COL15A1 COL4A1 COQ3 COQ5 COQ9 COX6A1 COX7B COX8A
CPT2 CRAT CS CYC1 CYP4B1 DBT DDT DECR1
DGAT1 DHCR7 DHRS7 DHRS7B DLAT DLD DNAJB9
DNAJC15 DRAM2 ECH1 ECHS1 ELMOD3 ELOVL6 ENPP2 EPHX2 ESRRA
ESYT1 ETFB FABP4 FAH FZD4 G3BP2 GADD45A GBE1
GHITM GPAM GPAT4 GPD2 GPHN GPX3 GPX4 GRPEL1
HADH HIBCH HSPB8 IDH1 IDH3A IDH3G IFNGR1 IMMT
ITGA7 ITIH5 ITSN1 JAGN1 LAMA4 LEP LIFR LIPE
LPCAT3 LPL LTC4S MAP4K3 MCCC1 MDH2 ME1 MGLL
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PIM3 PLIN2 POR PPARG PPM1B PPP1R15B PRDX3
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RETSAT RIOK3 RMDN3 RNF11 RREB1 RTN3 SAMM50 SCARB1
SCP2 SDHB SDHC SLC19A1 SLC1A5 SLC25A1 SLC25A10
SLC27A1 SLC5A6 SLC66A3 SNCG SOD1 SORBS1 SOWAHC
SPARCL1 SQOR SSPN STAT5A STOM SUCLG1 SULT1A1 TALDO1 TANK
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BCL3 BRCA1 C2 CAPG CARTPT CCL11 CCL13 CCL19
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CCR2 CCR5 CD1D CD2 CD247 CD28 CD3D CD3E
CD3G CD4 CD40 CD40LG CD47 CD7 CD74 CD79A
CD80 CD86 CD8A CD8B CD96 CDKN2A CFP CRTAM
CSF1 CSK CTSS CXCL13 CXCL9 CXCR3 DARS1 DEGS1
DYRK3 EGFR EIF3A EIF3D EIF3J EIF4G3 EIF5A ELANE
ELF4 EREG ETS1 F2 F2R FAS FASLG FCGR2B
FGR FLNA FYB1 GALNT1 GBP2 GCNT1 GLMN GPR65
GZMA GZMB HCLS1 HDAC9 HIF1A HLA-A HLA-DMA HLA-DMB
HLA-DOA HLA-DOB HLA-DQA1 HLA-DRA HLA-E HLA-G ICAM1
ICOSLG IFNAR2 IFNG IFNGR1 IFNGR2 IGSF6 IKBKB IL10
IL11 IL12A IL12B IL12RB1 IL13 IL15 IL16 IL18
IL18RAP IL1B IL2 IL27RA IL2RA IL2RB IL2RG IL4
IL4R IL6 IL7 IL9 INHBA INHBB IRF4 IRF7
IRF8 ITGAL ITGB2 ITK JAK2 KLRD1 KRT1 LCK
LCP2 LIF LTB LY75 LY86 LYN MAP3K7 MAP4K1
MBL2 MMP9 MRPL3 MTIF2 NCF4 NCK1 NCR1 NLRP3
NME1 NOS2 NPM1 PF4 PRF1 PRKCB PRKCG PSMB10
PTPN6 PTPRC RARS1 RIPK2 RPL39 RPL3L RPL9 RPS19

RPS3A RPS9 SIT1 SOCS1 SOCS5 SPI1 SRGN
ST8SIA4 STAB1 STAT1 STAT4 TAP1 TAP2 TAPBP TGFB1 TGFB2
THY1 TIMP1 TLR1 TLR2 TLR3 TLR6 TNF TPD52
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ELOVL5 FADS1 FKBP5 GNAI3 GPD1L GSR GUCY1A1 H1-0
HERC3 HMGCR HMGCS1 HOMER2 HPGD HSD17B14 IDI1
INPP4B INSIG1 IQGAP2 ITGAV KLK2 KLK3 KRT19 KRT8
LIFR LMAN1 MAF MAK MAP7 MERTK MYL12A NCOA4
NDRG1 NGLY1 NKX3-1 PA2G4 PDLIM5 PGM3 PIAS1 PLPP1
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SEC24D SELENOP SGK1 SLC26A2 SLC38A2 SMS SORD SPCS3
SPDEF SRF SRP19 STEAP4 STK39 TARP TMEM50A
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STC1 THBD TIMP1 TNFRSF21 VAV2 VCAN VEGFA
VTN
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ADAM9 ADAMTS5 ADRA1B AKT2 AKT3 ALOX15B AMH AMIGO1
AMIGO2 ARHGEF6 ARPC2 ATP1A3 B4GALT1 BAIAP2 BMP1 CADM2
CADM3 CALB2 CAP1 CD209 CD274 CD276 CD34 CD86
CD99 CDH1 CDH11 CDH15 CDH3 CDH4 CDH6 CDH8
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IKBKG INPPL1 INSIG1 IRS1 ITGA10 ITGA2 ITGA3 ITGA9
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MAPK13 MAPK14 MDK MMP2 MMP9 MPZL1 MPZL2 MSN
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NLGN3 NRAP NRTN NRXN2 PALS1 PARD6G PARVA PBX2
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SORBS3 SPEG SRC STX4 SYK SYMPK TAOK2 TGFBI

THBS3 THY1 TIAL1 TJP1 TMEM8B TNFRSF11B TRAF1
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PDGFRB PEA15 PLAT PLCB2 PLPPR4 PMAIP1 PPP2R5B PPP3R1
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RHOB RHOT2 RNASEL ROCK1 SAT1 SATB1 SC5D
SLC20A1 SMAD7 SOD1 SOD2 SPTAN1 SQSTM1 TAP1 TGFB2 TGFBR3
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BCAR3 BMP6 CAT CH25H CROT CYP27A1 CYP39A1
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HACL1 HAO1 HSD17B11 HSD17B4 HSD17B6 HSD3B1 HSD3B7
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MLYCD NEDD4 NPC1 NR0B2 NR1H4 NR1I2 NR3C2 NUDT12
OPTN PAOX PECR PEX1 PEX11A PEX11G PEX12 PEX13
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PNPLA8 PRDX5 PXMP2 RBP1 RETSAT RXRA RXRG SCP2
SERPINA6 SLC23A1 SLC23A2 SLC27A2 SLC27A5 SLC29A1
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CLU CPEB2 CTNNB1 CXCL16 CYP51A1 DHCR7 EBP ECH1
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CAPN5 CASP9 CD9 CFB CFD CFH CFI CLU
COMP CPB2 CPN1 CPQ CRIP2 CSRP1 CTSB CTSE
CTSH CTSK CTSO CTSV DCT DPP4 DUSP14 DUSP6
F10 F11 F12 F13B F2 F2RL2 F3 F8
F9 FBN1 FGA FGG FN1 FURIN FYN GDA
GNB2 GNG12 GP1BA GP9 GSN HMGCS2 HNF4A HPN
HRG HTRA1 ISCU ITGA2 ITGB3 ITIH1 KLF7 KLK8
KLKB1 LAMP2 LEFTY2 LGMN LRP1 LTA4H MAFF MASP2
MBL2 MEP1A MMP1 MMP10 MMP11 MMP14 MMP15 MMP2
MMP3 MMP7 MMP8 MMP9 MSRB2 MST1 OLR1 P2RY1
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PREP PROC PROS1 PROZ PRSS23 RABIF RAC1
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CASP5 CASP7 CASP9 CBLB CCL5 CD36 CD40LG CD46
CD55 CD59 CDA CDH13 CDK5R1 CEBPB CFB CFH
CLU COL4A2 CP CPM CPQ CR1 CR2 CSRP1
CTSB CTSC CTSD CTSH CTSL CTSO CTSS CTSV
CXCL1 DGKG DGKH DOCK10 DOCK4 DOCK9 DPP4 DUSP5
DUSP6 DYRK2 EHD1 ERAP2 F10 F2 F3 F5
F7 F8 FCER1G FCN1 FDX1 FN1 FYN GATA3
GCA GMFB GNAI2 GNAI3 GNB2 GNB4 GNG2 GNGT2
GP1BA GP9 GPD2 GRB2 GZMA GZMB GZMK HNF4A
HPCAL4 HSPA1A HSPA5 IL6 IRF1 IRF2 IRF7 ITGAM
ITIH1 JAK2 KCNIP2 KCNIP3 KIF2A KLK1 KLKB1 KYNU
L3MBTL4 LAMP2 LAP3 LCK LCP2 LGALS3 LGMN LIPA
LRP1 LTA4H LTF LYN MAFF ME1 MMP12 MMP13
MMP14 MMP15 MMP8 MSRB1 MT3 NOTCH4 OLR1 PCLO

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PIM1 PLA2G4A PLA2G7 PLAT PLAUR PLEK PLG PLSCR1
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SERPINB2 SERPINC1 SERPINE1 SERPING1 SH2B3
SIRT6 SPOCK2 SRC STX4 TFPI2 TIMP1 TIMP2
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CSTF3 DAD1 DCTN4 DDB1 DDB2 DGCR8 DGUOK DUT
EDF1 EIF1B ELL ELOA ERCC1 ERCC2 ERCC3 ERCC4
ERCC5 ERCC8 FEN1 GMPR2 GPX4 GSDME GTF2A2 GTF2B
GTF2F1 GTF2H1 GTF2H3 GTF2H5 GTF3C5 GUK1 HCLS1 HPRT1
IMPDH2 ITPA LIG1 MPC2 MPG MRPL40 NCBP2 NELFB
NELFCD NELFE NFX1 NME1 NME3 NME4 NPR2 NT5C
NT5C3A NUDT21 NUDT9 PCNA PDE4B PDE6G PNP POLA1
POLA2 POLB POLD1 POLD3 POLD4 POLE4 POLH POLL
POLR1C POLR1D POLR1H POLR2A POLR2C POLR2D POLR2E POLR2F
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PRIM1 RAD51 RAD52 RAE1 RALA RBX1 REV3L RFC2
RFC3 RFC4 RFC5 RNMT RPA2 RPA3 RRM2B SAC3D1
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STX3 SUPT4H1 SUPT5H SURF1 TAF10 TAF12 TAF13 TAF1C
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UMPS UPF3B USP11 VPS28 VPS37B VPS37D XPC ZNF707
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EED EIF2S1 ESPL1 EXOSC8 EZH2 GINS1 GINS3 GINS4
GSPT1 H2AX H2AZ1 HELLS HMGA1 HMGB2 HMGB3 HMMR
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KIF2C KIF4A KPNA2 LBR LIG1 LMNB1 LUC7L3 LYAR
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MKI67 MLH1 MMS22L MRE11 MSH2 MTHFD2 MXD3 MYBL2
MYC NAA38 NAP1L1 NASP NBN NCAPD2 NME1 NOLC1
NOP56 NUDT21 NUP107 NUP153 NUP205 ORC2 ORC6 PA2G4
PAICS PAN2 PCNA PDS5B PHF5A PLK1 PLK4 PMS2
PNN POLA2 POLD1 POLD2 POLD3 POLE POLE4 POP7
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CXCL1 CXCL12 CXCL6 CXCL8 DAB2 DCN DKK1 DPYSL3
DST ECM1 ECM2 EDIL3 EFEMP2 ELN EMP3 ENO2
FAP FAS FBLN1 FBLN2 FBLN5 FBN1 FBN2 FERMT2
FGF2 FLNA FMOD FN1 FOXC2 FSTL1 FSTL3 FUCA1
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LOX LOXL1 LOXL2 LRP1 LRRC15 LUM MAGEE1 MATN2
MATN3 MCM7 MEST MFAP5 MGP MMP1 MMP14 MMP2
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NT5E NTM OXTR P3H1 PCOLCE PCOLCE2 PDGFRB PDLIM4
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PPIB PRRX1 PRSS2 PTHLH PTX3 PVR QSOX1 RGS4
RHOB SAT1 SCG2 SDC1 SDC4 SERPINE1 SERPINE2 SERPINH1 SFRP1 SFRP4 SGCB SGCD SGCG
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TAGLN TFPI2 TGFB1 TGFBI TGFBR3 TGM2 THBS1 THBS2
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FHL2 FKBP4 FKBP5 FLNB FOS FOXC1 FRK GAB2
GFRA1 GJA1 GLA GREB1 HES1 HR HSPB8 IGF1R
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MUC1 MYB MYBBP1A MYBL1 MYC MYOF NADSYN1 NAV2
NBL1 NCOR2 NHERF1 NPY1R NRIP1 NXT1 OLFM1 OLFML3
OPN3 OVOL2 P2RY2 PAPSS2 PDLIM3 PDZK1 PEX11A PGR
PLAAT3 PMAIP1 PODXL PPIF PRSS23 PTGES RAB17 RAB31
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SNX24 SOX3 STC2 SULT2B1 SVIL SYBU SYNGR1 SYT12
TBC1D30 TFAP2C TFF1 TFF3 TGIF2 TGM2 THSD4 TIAM1
TIPARP TJP3 TMEM164 TMPRSS3 TOB1 TPBG TPD52L1 TSKU
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CELSR2 CHPT1 CHST8 CISH CKB CLIC3 COX6C CPE
CXCL12 CXCL14 CYP26B1 CYP4F11 DCXR DHCR7 DHRS2 DLG5
DNAJC1 DNAJC12 DUSP2 DYNLT3 EEIG1 EGR3 ELOVL5 EMP2
ETFB FABP5 FARP1 FDFT1 FGFR3 FKBP4 FKBP5 FLNB
FOS FOXC1 FRK GAL GALE GFUS GINS2 GJB3
GLA GPER1 HMGCS2 HOMER2 HPRT1 HR HSPA4L HSPB8
ID2 IDH2 IGFBP4 IGSF1 IL17RB IL6ST IMPA2 ISG20
ITPK1 JAK1 JAK2 KCNK5 KIF20A KLF4 KLK10 KLK11
KRT13 KRT19 LAMC2 LARGE1 LLGL2 LSR LTF MAPK13
MAPT MDK MEST METTL3 MICB MOCS2 MYB MYOF
NAB2 NBL1 NCOR2 NHERF1 NMU NPY1R NRIP1 NXT1
OLFM1 OPN3 OVOL2 PAPSS2 PCP4 PDCD4 PDLIM3 PDZK1
PERP PGR PKP3 PLAAT3 PLAC1 PLK4 PLXNB1 PPIF
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ALDH9A1 ALDOA AOC3 APEX1 AQP7 AUH BCKDHB BLVRA BMPR1B
BPHL CA2 CA4 CA6 CBR1 CBR3 CD1D CD36
CEL CIDEA CPOX CPT1A CPT2 CRAT CRYZ CYP1A1
CYP4A11 CYP4A22 D2HGDH DECR1 DHCR24 DLD DLST ECH1

ECHS1 ECI1 ECI2 EHHADH ELOVL5 ENO2 ENO3 EPHX1
ERP29 ETFDH FABP1 FABP2 FASN FH FMO1 G0S2
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GRHPR GSTZ1 H2AZ1 HADH HADHB HAO2 HCCS HIBCH
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MAOA MCEE MDH1 MDH2 ME1 METAP1 MGLL MIF
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PCBD1 PDHA1 PDHB PPARA PRDX6 PSME1 PTPRG PTS
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CHST12 CHST2 CHST4 CHST6 CITED2 CLDN3 CLDN9 CLN6
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ISG20 KDELR3 KIF20A KIF2A LCT LDHA LDHC LHPP
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IL3RA IL4R IRF4 IRF6 IRF8 ITGA6 ITGAE ITGAV
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IFIT2 IFIT3 IFITM1 IFITM2 IFITM3 IL15 IL4R IL7
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ATG10 AVL9 BIRC3 BMP2 BPGM BTBD3 BTC C3AR1
CA2 CAB39L CBL CBR4 CBX8 CCL20 CCND2 CCSER2
CD37 CDADC1 CFB CFH CFHR2 CIDEA CLEC4A CMKLR1
CPE CROT CSF2 CSF2RA CTSS CXCL10 CXCR4 DCBLD2
DNMBP DOCK2 DUSP6 EMP1 ENG EPB41L3 EPHB2 EREG
ERO1A ETS1 ETV1 ETV4 ETV5 EVI5 F13A1 F2RL1
FBXO4 FCER1G FGF9 FLT4 FUCA1 G0S2 GABRA3
GADD45G GALNT3 GFPT2 GLRX GNG11 GPNMB GPRC5B GUCY1A1 GYPC
H2BC3 HBEGF HDAC9 HKDC1 HOXD11 HSD11B1 ID2 IGF2
IGFBP3 IKZF1 IL10RA IL1B IL1RL2 IL2RG IL33 IL7R
INHBA IRF8 ITGA2 ITGB2 ITGBL1 JUP KCNN4 KIF5C
KLF4 LAPTM5 LAT2 LCP1 LIF LY96 MAFB MALL
MAP3K1 MAP4K1 MAP7 MMD MMP10 MMP11 MMP9 MPZL2
MTMR10 MYCN NAP1L2 NGF NIN NR0B2 NR1H4 NRP1
PCP4 PCSK1N PDCD1LG2 PECAM1 PEG3 PIGR PLAT
PLAU PLAUR PLEK2 PLVAP PPBP PPP1R15A PRDM1
PRELID3B PRKG2 PRRX1 PSMB8 PTBP2 PTCD2 PTGS2
PTPRR RABGAP1L RBM4 RBP4 RELN RETN RGS16
SATB1 SCG3 SCG5 SCN1B SDCCAG8 SEMA3B SERPINA3
SLPI SNAP25 SNAP91 SOX9 SPARCL1 SPON1 SPP1 SPRY2
ST6GAL1 STRN TFPI TLR8 TMEM100 TMEM158 TMEM176A
TMEM176B TNFAIP3 TNFRSF1B TNNT2 TOR1AIP2
TPH1 TRAF1 TRIB1 TRIB2 TSPAN1 TSPAN13 TSPAN7 USH1C
USP12 VWA5A WDR33 WNT7A YRDC ZNF277 ZNF639
HALLMARK_MITOTIC_SPINDLE https://www.gsea-msigdb.org/gsea/msigdb/human/geneset/HALLMARK_MITOTIC_SPINDLE ABI1 ABL1 ABR

ACTN4 AKAP13 ALMS1 ALS2 ANLN APC ARAP3 ARF6
ARFGEF1 ARFIP2 ARHGAP10 ARHGAP27 ARHGAP29
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BIN1 BIRC5 BRCA2 BUB1 CAPZB CCDC88A CCNB2 CD2AP
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CDC42EP4 CDK1 CDK5RAP2 CENPE CENPF CEP131
CEP192 CEP250 CEP57 CEP72 CKAP5 CLASP1 CLIP1 CLIP2
CNTRL CNTROB CPAP CSNK1D CTTN CYTH2 DLG1 DLGAP5
DOCK2 DOCK4 DST DYNC1H1 DYNLL2 ECT2 EPB41
EPB41L2 ESPL1 EZR FARP1 FBXO5 FGD4 FGD6 FLNA FLNB
FSCN1 GEMIN4 GSN HDAC6 HOOK3 INCENP ITSN1 KATNA1
KATNB1 KIF11 KIF15 KIF1B KIF20B KIF22 KIF23 KIF2C
KIF3B KIF3C KIF4A KIF5B KIFAP3 KLC1 KNTC1 KPTN
LATS1 LLGL1 LMNB1 LRPPRC MAP1S MAP3K11 MAPRE1 MARCKS
MARK4 MID1 MID1IP1 MYH10 MYH9 MYO1E MYO9B NCK1
NCK2 NDC80 NEDD9 NEK2 NET1 NF1 NIN NOTCH2
NUMA1 NUSAP1 OPHN1 PAFAH1B1 PALLD PCGF5 PCM1
PCNT PDLIM5 PIF1 PKD2 PLEKHG2 PLK1 PPP4R2 PRC1
PREX1 PXN RAB3GAP1 RABGAP1 RACGAP1 RALBP1 RANBP9
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RICTOR ROCK1 SAC3D1 SASS6 SEPTIN9 SHROOM1 SHROOM2 SMC1A
SMC3 SMC4 SORBS2 SOS1 SPTAN1 SPTBN1 SSH2 STAU1
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CYB5B CYP51A1 DAPP1 DDIT3 DDIT4 DDX39A DHCR24 DHCR7
DHFR EBP EDEM1 EEF1E1 EGLN3 EIF2S2 ELOVL5 ELOVL6
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FKBP2 G6PD GAPDH GBE1 GCLC GGA2 GLA GLRX
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ITGB2 LDHA LDLR LGMN LTA4H M6PR MAP2K3 MCM2
MCM4 ME1 MLLT11 MTHFD2 MTHFD2L NAMPT NFIL3 NFKBIB
NFYC NHERF1 NIBAN1 NMT1 NUFIP1 NUP205 NUPR1 P4HA1
PDAP1 PDK1 PFKL PGK1 PGM1 PHGDH PIK3R3 PITPNB
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PPP1R15A PRDX1 PSAT1 PSMA3 PSMA4 PSMB5 PSMC2 PSMC4
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RAB1A RDH11 RIT1 RPA1 RPN1 RRM2 RRP9 SC5D
SCD SDF2L1 SEC11A SERP1 SERPINH1 SHMT2 SKAP2
SLA SLC1A4 SLC1A5 SLC2A1 SLC2A3 SLC37A4 SLC6A6
SLC7A11 SLC7A5 SORD SQLE SQSTM1 SRD5A1 SSR1 STARD4 STC1

STIP1 SYTL2 TBK1 TCEA1 TES TFRC TM7SF2 TMEM97
TOMM40 TPI1 TRIB3 TUBA4A TUBG1 TXNRD1 UBE2D3 UCHL5
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CNBP COPS5 COX5A CSTF2 CTPS1 CUL1 CYC1 DDX18
DDX21 DEK DHX15 DUT EEF1B2 EIF1AX EIF2S1 EIF2S2
EIF3B EIF3D EIF3J EIF4A1 EIF4E EIF4G2 EIF4H EPRS1
ERH ETF1 EXOSC7 FAM120A FBL G3BP1 GLO1 GNL3
GOT2 GSPT1 H2AZ1 HDAC2 HDDC2 HDGF HNRNPA1
HNRNPA2B1 HNRNPA3 HNRNPC HNRNPD HNRNPR HNRNPU HPRT1
HSP90AB1 HSPD1 HSPE1 IARS1 IFRD1 ILF2 IMPDH2 KARS1
KPNA2 KPNB1 LDHA LSM2 LSM7 MAD2L1 MCM2 MCM4
MCM5 MCM6 MCM7 MRPL23 MRPL9 MRPS18B MYC NAP1L1
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PGK1 PHB1 PHB2 POLD2 POLE3 PPIA PPM1G PRDX3
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PSMB2 PSMB3 PSMC4 PSMC6 PSMD1 PSMD14 PSMD3 PSMD7
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RNPS1 RPL14 RPL18 RPL22 RPL34 RPL6 RPLP0 RPS10
RPS2 RPS3 RPS5 RPS6 RRM1 RRP9 RSL1D1 RUVBL2
SERBP1 SET SF3A1 SF3B3 SLC25A3 SMARCC1 SNRPA SNRPA1
SNRPB2 SNRPD1 SNRPD2 SNRPD3 SNRPG SRM SRPK1 SRSF1
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TCP1 TFDP1 TOMM70 TRA2B TRIM28 TUFM TXNL4A TYMS
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DDX18 DUSP2 EXOSC5 FARSA GNL3 GRWD1 HK2 HSPD1
HSPE1 IMP4 IPO4 LAS1L MAP3K6 MCM4 MCM5
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NOP16 NOP2 NOP56 NPM1 PA2G4 PES1 PHB1 PLK1
PLK4 PPAN PPRC1 PRMT3 PUS1 RABEPK RCL1 RRP12
RRP9 SLC19A1 SLC29A2 SORD SRM SUPV3L1 TBRG4 TCOF1
TFB2M TMEM97 UNG UTP20 WDR43 WDR74
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CKB CKM CKMT2 CLU CNN3 COL15A1 COL1A1 COL3A1
COL4A2 COL6A2 COL6A3 COX6A2 COX7A1 CRAT CRYAB CSRP3
CTF1 DAPK2 DENND2B DES DMD DMPK DTNA EFS
EIF4A2 ENO3 EPHB3 ERBB3 FABP3 FDPS FGF2 FHL1
FKBP1B FLII FOXO4 FST FXYD1 GAA GABARAPL2

GADD45B GJA5 GNAO1 GPX3 GSN HBEGF HDAC5 HRC
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ITGB4 ITGB5 KCNH1 KCNH2 KIFC3 KLF5 LAMA2 LARGE1
LDB3 LPIN1 LSP1 MAPK12 MAPRE3 MB MEF2A MEF2C
MEF2D MRAS MYBPC3 MYBPH MYF6 MYH1 MYH11 MYH2
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MYOM2 MYOZ1 NAV2 NCAM1 NOS1 NOTCH1 NQO1 OCEL1
PC PDE4DIP PDLIM7 PFKM PGAM2 PICK1 PKIA PLXNB2
PPFIA4 PPP1R3C PRNP PSEN2 PTGIS PTP4A3 PVALB PYGM
RB1 REEP1 RIT1 RYR1 SCD SCHIP1 SGCA SGCD
SGCG SH2B1 SH3BGR SIRT2 SLC6A8 SLN SMTN SOD3
SORBS1 SORBS3 SPARC SPDEF SPEG SPHK1 SPTAN1 SSPN
STC2 SVIL SYNGR2 TAGLN TCAP TEAD4 TGFB1 TNNC1
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FZD7 HES1 HEYL JAG1 KAT2A LFNG MAML2 NOTCH1
NOTCH2 NOTCH3 PPARD PRKCA PSEN2 PSENEN RBX1 SAP30
SKP1 ST3GAL6 TCF7L2 WNT2 WNT5A
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ATP5F1D ATP5F1E ATP5MC1 ATP5MC2 ATP5MC3 ATP5ME ATP5MF ATP5MG ATP5PB
ATP5PD ATP5PF ATP5PO ATP6AP1 ATP6V0B ATP6V0C ATP6V0E1
ATP6V1C1 ATP6V1D ATP6V1E1 ATP6V1F ATP6V1G1
ATP6V1H BAX BCKDHA BDH2 CASP7 COX10 COX11 COX15
COX17 COX4I1 COX5A COX5B COX6A1 COX6B1 COX6C COX7A2
COX7A2L COX7B COX7C COX8A CPT1A CS CYB5A CYB5R3
CYC1 CYCS DECR1 DLAT DLD DLST ECH1 ECHS1
ECI1 ETFA ETFB ETFDH FDX1 FH FXN GLUD1
GOT2 GPI GPX4 GRPEL1 HADHA HADHB HCCS
HSD17B10 HSPA9 HTRA2 IDH1 IDH2 IDH3A IDH3B IDH3G
IMMT ISCA1 ISCU LDHA LDHB LRPPRC MAOB MDH1
MDH2 MFN2 MGST3 MPC1 MRPL11 MRPL15 MRPL34 MRPL35
MRPS11 MRPS12 MRPS15 MRPS22 MRPS30 MTRF1 MTRR MTX2
NDUFA1 NDUFA2 NDUFA3 NDUFA4 NDUFA5 NDUFA6 NDUFA7 NDUFA8
NDUFA9 NDUFAB1 NDUFB1 NDUFB2 NDUFB3 NDUFB4 NDUFB5 NDUFB6
NDUFB7 NDUFB8 NDUFC1 NDUFC2 NDUFS1 NDUFS2 NDUFS3 NDUFS4
NDUFS6 NDUFS7 NDUFS8 NDUFV1 NDUFV2 NNT NQO2 OAT
OGDH OPA1 OXA1L PDHA1 PDHB PDHX PDK4 PDP1
PHB2 PHYH PMPCA POLR2F POR PRDX3 RETSAT RHOT1
RHOT2 SDHA SDHB SDHC SDHD SLC25A11
SLC25A12 SLC25A20 SLC25A3 SLC25A4 SLC25A5 SLC25A6 SUCLA2
SUCLG1 SUPV3L1 SURF1 TCIRG1 TIMM10 TIMM13 TIMM17A TIMM50
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FGF13 FOS FOXO3 FUCA1 GADD45A GLS2 GM2A GPX2
H1-2 H2AC25 H2AJ HBEGF HDAC3 HEXIM1 HINT1 HMOX1
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IRAG2 IRAK1 ISCU ITGB4 JAG2 JUN KIF13B KLF4
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RRP8 RXRA S100A10 S100A4 SAT1 SDC1 SEC61A1
SERPINB5 SERTAD3 SESN1 SFN SLC19A2 SLC35D1 SLC3A2
SLC7A11 SOCS1 SP1 SPHK1 ST14 STEAP3 STOM TAP1
TAX1BP3 TCHH TCN2 TGFA TGFB1 TM4SF1 TM7SF3 TNFSF9 TNNI1
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GCK HNF1A IAPP INS INSM1 ISL1 LMO2 MAFB
NEUROD1 NEUROG3 NKX2-2 NKX6-1 PAK3 PAX4 PAX6 PCSK1
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ECI2 EHHADH ELOVL5 EPHX2 ERCC1 ERCC3 ESR2 FABP6
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ACTR2 ACTR3 ADCY2 AKT1 AKT1S1 AP2M1 ARF1
ARHGDIA ARPC3 ATF1 CAB39 CAB39L CALR CAMK4 CDK1 CDK2
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FGF22 FGF6 GNA14 GNGT1 GRB2 GRK2 GSK3B HRAS
HSP90B1 IL2RG IL4 IRAK4 ITPR2 LCK MAP2K3 MAP2K6
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MYD88 NCK1 NFKBIB NGF NOD1 PAK4 PDK1 PFN1
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PPP1CA PPP2R1B PRKAA2 PRKAG1 PRKAR2A PRKCB PTEN PTPN11
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TIAM1 TNFRSF1A TRAF2 TRIB3 TSC2 UBE2D3 UBE2N
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SCRN1 SEC22B SEC24D SEC31A SGMS1 SH3GL2 SNAP23 SNX2
SOD1 SSPN STAM STX12 STX16 STX7 TMED10 TMED2
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VAMP7 VPS45 VPS4B YIPF6 YKT6 ZW10
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msigdb.org/gsea/msigdb/human/geneset/
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GAPDHS GFI1 GMCL1 GPR182 GRM8 GSG1 GSTM3 H1-6
HBZ HOXB1 HSPA1L HSPA2 HSPA4L HTR5A IDE IFT88
IL12RB2 IL13RA2 IP6K1 JAM3 KIF2C LDHC LPIN1 MAP7
MAST2 MEP1B MLF1 MLLT10 MTNR1A MTOR NAA11 NCAPH
NEFH NEK2 NF2 NOS1 NPHP1 NPY5R OAZ3 ODF1

PACRG PAPOLB PARP2 PCSK1N PCSK4 PDHA2 PEBP1 PGK2
PGS1 PHF7 PHKG2 PIAS2 POMC PRKAR2A PRM2 PSMG1
RAD17 RFC4 RPL39L SCG3 SCG5 SEPTIN4 SHE SIRT1
SLC12A2 SLC2A5 SNAP91 SPATA6 SPMAP2 STAM2 STRBP SYCP1
TALDO1 TCP11 TEKT2 TKTL1 TLE4 TNNI3 TNP1 TNP2
TOPBP1 TSN TSSK2 TTK TUBA3C TULP2 VDAC3 YBX2
ZC2HC1C ZC3H14 ZNRF4 ZPBP
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CTNNB1 ENG FKBP1A FNTA FURIN HDAC1 HIPK2 ID1
ID2 ID3 IFNGR2 JUNB KLF10 LEFTY2 LTBP2 MAP3K7
NCOR2 NOG PMEPA1 PPM1A PPP1CA PPP1R15A RAB31
RHOA SERPINE1 SKI SKIL SLC20A1 SMAD1 SMAD3
SMAD6 SMAD7 SMURF1 SMURF2 SPTBN1 TGFB1 TGFBR1 TGIF1
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ACKR3 AREG ATF3 ATP2B1 B4GALT1 B4GALT5 BCL2A1 BCL3
BCL6 BHLHE40 BIRC2 BIRC3 BMP2 BTG1 BTG2 BTG3
CCL2 CCL20 CCL4 CCL5 CCN1 CCND1 CCNL1 CCRL2
CD44 CD69 CD80 CD83 CDKN1A CEBPB CEBPD CFLAR
CLCF1 CSF1 CSF2 CXCL1 CXCL10 CXCL11 CXCL2 CXCL3
CXCL6 DENND5A DNAJB4 DRAM1 DUSP1 DUSP2 DUSP4 DUSP5
EDN1 EFNA1 EGR1 EGR2 EGR3 EHD1 EIF1 ETS2
F2RL1 F3 FJX1 FOS FOSB FOSL1 FOSL2 FUT4
G0S2 GADD45A GADD45B GCH1 GEM GFPT2 GPR183 HBEGF
HES1 ICAM1 ICOSLG ID2 IER2 IER3 IER5 IFIH1
IFIT2 IFNGR2 IL12B IL15RA IL18 IL1A IL1B IL23A
IL6 IL6ST IL7R INHBA IRF1 IRS2 JAG1 JUN
JUNB KDM6B KLF10 KLF2 KLF4 KLF6 KLF9 KYNU
LAMB3 LDLR LIF LITAF MAFF MAP2K3 MAP3K8 MARCKS
MCL1 MSC MXD1 MYC NAMPT NFAT5 NFE2L2 NFIL3
NFKB1 NFKB2 NFKBIA NFKBIE NINJ1 NR4A1 NR4A2 NR4A3
OLR1 PANX1 PDE4B PDLIM5 PER1 PFKFB3 PHLDA1 PHLDA2
PLAU PLAUR PLEK PLK2 PLPP3 PMEPA1 PNRC1
PPP1R15A PTGER4 PTGS2 PTPRE PTX3 RCAN1 REL RELA
RELB RHOB RIGI RIPK2 RNF19B SAT1 SDC4
SERPINB2 SERPINB8 SERPINE1 SGK1 SIK1
SLC16A6 SLC2A3 SLC2A6 SMAD3 SNN SOCS3 SOD2 SPHK1 SPSB1
SQSTM1 STAT5A TANK TAP1 TGIF1 TIPARP TLR2 TNC
TNF TNFAIP2 TNFAIP3 TNFAIP6 TNFAIP8 TNFRSF9 TNFSF9 TNIP1
TNIP2 TRAF1 TRIB1 TRIP10 TSC22D1 TUBB2A VEGFA YRDC
ZBTB10 ZC3H12A ZFP36
HALLMARK_UNFOLDED_PROTEIN_RESPONSE https://www.gsea-msigdb.org/gsea/msigdb/human/geneset/HALLMARK_UNFOLDED_PROTEIN_RESPONSE
ALDH18A1 ARFGAP1 ASNS ATF3 ATF4 ATF6 ATP6V0D1
BAG3 BANF1 CALR CCL2 CEBPB CEBPG CHAC1 CKS1B
CNOT2 CNOT4 CNOT6 CXXC1 DCP1A DCP2 DCTN1 DDIT4
DDX10 DKC1 DNAJA4 DNAJB9 DNAJC3 EDC4 EDEM1 EEF2

EIF2AK3 EIF2S1 EIF4A1 EIF4A2 EIF4A3 EIF4E EIF4EBP1
EIF4G1 ERN1 ERO1A EXOC2 EXOSC1 EXOSC10 EXOSC2 EXOSC4
EXOSC5 EXOSC9 FKBP14 FUS GEMIN4 GOSR2 H2AX
HERPUD1 HSP90B1 HSPA5 HSPA9 HYOU1 IARS1 IFIT1 IGFBP1 IMP3
KDELR3 KHSRP KIF5B LSM1 LSM4 MTHFD2 MTREX NABP1
NFYA NFYB NHP2 NOLC1 NOP14 NOP56 NPM1 PAIP1
PARN PDIA5 PDIA6 POP4 PREB PSAT1 RPS14 RRP9
SDAD1 SEC11A SEC31A SERP1 SHC1 SKIC3 SLC1A4
SLC30A5 SLC7A5 SPCS1 SPCS3 SRPRA SRPRB SSR1 STC2 TARS1
TATDN2 TSPYL2 TUBB2A VEGFA WFS1 WIPI1 XBP1 XPOT
YIF1A YWHAZ ZBTB17
HALLMARK_UV_RESPONSE_DN https://www.gsea-msigdb.org/gsea/msigdb/human/geneset/HALLMARK_UV_RESPONSE_DN ABCC1 ACVR2A ADD3 ADGRL2
ADORA2B AGGF1 AKT3 AMPH ANXA2 ANXA4 APBB2 ARHGEF9 ATP2B1
ATP2B4 ATP2C1 ATRN ATRX ATXN1 BCKDHB BDNF
BHLHE40 BMPR1A CACNA1A CAP2 CAV1 CCN1 CDC42BPA CDK13
CDKN1B CDON CELF2 CITED2 COL11A1 COL1A1 COL1A2 COL3A1
COL5A2 DAB2 DBP DDAH1 DLC1 DLG1 DMAC2L DUSP1
DYRK1A EFEMP1 ERBB2 F3 FBLN5 FHL2 FYN FZD2
GCNT1 GJA1 GRK5 HAS2 ICA1 ID1 IGF1R IGFBP5
INPP4B INSIG1 IRS1 ITGB3 KALRN KCNMA1 KIT LAMC1
LDLR LPAR1 LTBP1 MAGI2 MAP1B MAP2K5 MAPK14 MET
MGLL MGMT MIOS MMP16 MRPS31 MT1E MTA1 MYC
NEK7 NFIB NFKB1 NIPBL NOTCH2 NR1D2 NR3C1 NRP1
PDGFRB PDLIM5 PEX14 PHF3 PIAS3 PIK3CD PIK3R3 PLCB4
PLPP3 PMP22 PPARG PRDM2 PRKAR2B PRKCA PRKCE PTEN
PTGFR PTPN21 PTPRM RASA2 RBPMS RGS4 RND3 RUNX1
RXRA SCAF8 SCHIP1 SCN8A SDC2 SERPINE1 SFMBT1
SIPA1L1 SLC67A1 SLC7A1 SMAD3 SMAD7 SNAI2 SPOP SRI
SYNE1 SYNJ2 TENT4A TFPI TGFBR2 TGFBR3 TJP1
TOGARAM1 VAV2 VLDLR WDR37 YTHDC1 ZMIZ1
HALLMARK_UV_RESPONSE_UP https://www.gsea-msigdb.org/gsea/msigdb/human/geneset/HALLMARK_UV_RESPONSE_UP ABCB1 ACAA1 AGO2 ALAS1 ALDOA
AMD1 AP2S1 APOM AQP3 ARRB2 ASNS ATF3
ATP6V1C1 ATP6V1F BAK1 BCL2L11 BID BMP2 BSG BTG1
BTG2 BTG3 C4BPB CA2 CASP3 CCK CCND3 CCNE1
CDC34 CDC5L CDK2 CDKN1C CDKN2B CDO1 CEBPG CHKA
CHRNA5 CLCN2 CLTB CNP COL2A1 CREG1 CTSV CXCL2
CYB5B CYB5R1 CYP1A1 DDX21 DGAT1 DLG4 DNAJA1 DNAJB1
E2F5 EIF2S3 EIF5 ENO2 EPCAM EPHX1 FEN1 FGF18
FKBP4 FMO1 FOS FOSB FURIN GAL GCH1 GGH
GLS GPX3 GRINA GRPEL1 H2AX HLA-F HMOX1 HNRNPU
HSPA13 HSPA2 HTR7 HYAL2 ICAM1 IGFBP2 IL6 IL6ST
IRF1 JUNB KCNH2 KLHDC3 LHX2 LYN MAOA
MAPK8IP2 MARK2 MGAT1 MMP14 MRPL23 MSX1 NAT1 NFKBIA
NKX2-5 NPTX2 NPTXR NR4A1 NTRK3 NUP58 NXF1 OLFM1
ONECUT1 PARP2 PDAP1 PDLIM3 PLCL1 POLE3 POLG2 POLR2H
PPAT PPIF PPP1R2 PPT1 PRKACA PRKCD PRPF3 PSMC3
PTPRD RAB27A RASGRP1 RET RFC4 RHOB RPN1 RRAD
RXRB SELENOW SHOX2 SIGMAR1 SLC25A4 SLC6A12 SLC6A8 SOD2

SPR SQSTM1 STARD3 STIP1 STK25 SULT1A1 TACR3 TAP1
TARS1 TCHH TFRC TGFBRAP1 TMBIM6 TST TUBA4A
TYRO3 UROD WIZ YKT6
HALLMARK_WNT_BETA_CATENIN_SIGNALING https://www.gsea-msigdb.org/gsea/msigdb/human/geneset/HALLMARK_WNT_BETA_CATENIN_SIGNALING ADAM17
AXIN1 AXIN2 CCND2 CSNK1E CTNNB1 CUL1 DKK1 DKK4
DLL1 DVL2 FRAT1 FZD1 FZD8 GNAI1 HDAC11 HDAC2
HDAC5 HEY1 HEY2 JAG1 JAG2 KAT2A LEF1 MAML1
MYC NCOR2 NCSTN NKD1 NOTCH1 NOTCH4 NUMB PPARD
PSEN2 PTCH1 RBPJ SKP2 TCF7 TP53 WNT1 WNT5B
WNT6
HALLMARK_XENOBIOTIC_METABOLISM https://www.gsea-msigdb.org/gsea/msigdb/human/geneset/HALLMARK_XENOBIOTIC_METABOLISM ABCC2 ABCC3
ABCD2 ABHD6 ACO2 ACOX1 ACOX2 ACOX3 ACP1 ACP2
ACSM1 ADH1C ADH5 ADH7 AHCY AKR1C2 AKR1C3 ALAS1
ALDH2 ALDH3A1 ALDH9A1 ANGPTL3 AOX1 AP4B1 APOE AQP9
ARG1 ARG2 ARPP19 ASL ATOH8 ATP2A2 BCAR1 BCAT1
BLVRB BPHL CA2 CASP6 CAT CBR1 CCL25 CD36
CDA CDO1 CES1 CFB CNDP2 COMT CROT CRP
CSAD CYB5A CYFIP2 CYP17A1 CYP1A1 CYP1A2 CYP26A1
CYP27A1 CYP2C18 CYP2E1 CYP2J2 CYP2S1 CYP4F2 DCXR DDAH2 DDC
DDT DHPS DHRS1 DHRS7 ECH1 ELOVL5 ENPEP ENTPD5
EPHA2 EPHX1 ESR1 ETFDH ETS2 F10 F11 FABP1
FAH FAS FBLN1 FBP1 FETUB FMO1 FMO3 G6PC1
GABARAPL1 GAD1 GART GCH1 GCKR GCLC GCNT2
GNMT GSR GSS GSTA3 GSTM4 GSTO1 GSTT2 HACL1
HES6 HGFAC HMOX1 HNF4A HPRT1 HRG HSD11B1
HSD17B2 ID2 IDH1 IGF1 IGFBP1 IGFBP4 IL1R1 IRF8 ITIH1
ITIH4 JUP KARS1 KYNU LCAT LEAP2 LONP1 LPIN2
MAN1A1 MAOA MARCHF6 MBL2 MCCC2 MPP2 MT2A MTHFD1
NDRG2 NFS1 NINJ1 NMT1 NPC1 NQO1 PAPSS2 PC
PDK4 PDLIM5 PEMT PGD PGRMC1 PINK1 PLG PMM1
POR PPARD PROS1 PSMB10 PTGDS PTGES PTGES3 PTGR1
PTS PYCR1 RAP1GAP RBP4 REG1A RETSAT SAR1B
SERPINA6 SERPINE1 SERTAD1 SHMT2 SLC12A4 SLC1A5
SLC22A1 SLC35B1 SLC35D1 SLC46A3 SLC6A12 SLC6A6 SMOX SPINT2 SSR3
TAT TDO2 TGFB2 TKFC TMBIM6 TMEM176B TMEM97
TNFRSF1A TPST1 TTPA TYR UGDH UPB1 UPP1
VNN1 VTN XDH